

Volumes by the Disc and Washer Methods



The disc method (about the x-axis)

- 1 Find the volume of the solid formed by revolving $y = \sqrt{x}$ about the x-axis on $[0, 4]$.
 - 2 Find the volume of the solid formed by revolving $y = x^2$ about the x-axis on $[0, 2]$.
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The disc method (about the y-axis)

- 3 Find the volume formed by revolving $x = y^2$ about the y-axis on $y \in [0, 2]$.
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The washer method

- 4 Find the volume formed by revolving the region between $y = x$ and $y = x^2$ (on $[0, 1]$) about the x-axis.
 - 5 Find the volume formed by revolving the region between $y = \sqrt{x}$ and $y = x$ about the x-axis on $[0, 1]$.
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Volumes about a horizontal or vertical line (not an axis)

- 6 Find the volume formed by revolving the region bounded by $y = x^2$ and $y = 4$ about the line $y = 4$.
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Volumes with known cross-sections

- 7 The base of a solid is the region between $y = x^2$ and $y = 4$ on $[-2, 2]$. Cross-sections perpendicular to the x-axis are squares. Find the volume.
 - 8 The base of a solid is a circle of radius 2 ($x^2 + y^2 = 4$). Cross-sections perpendicular to the x-axis are semicircles. Find the volume.
 - 9 Set up (but do not evaluate) the integral for the volume of the solid whose base is the region bounded by $y = \sin x$ and the x-axis on $[0, \pi]$, with cross-sections perpendicular to the x-axis that are equilateral triangles.
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Volume by revolving about the y-axis (washer method)

- 10 Find the volume formed by revolving the region between $x = y^2$ and $x = 2y$ about the y -axis.